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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Link
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 4: Quality Education SDG 8: Decent Work and Economic Growth SDG 9: Industry, Innovation and Infrastructure
Project / Solution Name	Link - AI-Powered Onboarding for Teams
Name (Leader)	Samuel Phang Han Wen
Email & Phone (Leader)	ait2404058@xmu.edu.my / +60 11-2008 8045
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and medium-sized businesses often rely on experienced staff to explain procedures to each new hire. Important knowledge may be spread across documents or passed on verbally, so employees can struggle to find consistent, current answers while supervisors repeat the same guidance. This makes onboarding harder to scale for teams with limited training capacity. The OECD's 2025 D4SME Survey found that 13% of surveyed SMEs used formal training to build digital skills; 35% relied on internet research and 23% on colleague knowledge-sharing. This is evidence of a broader skills-access gap, not a Link-specific outcome. Link addresses the challenge by organizing approved company knowledge into guided learning and source-cited question answering. A pilot is needed to measure completion and time-to-completion against each team's existing process. No productivity or environmental improvement has yet been measured.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Link is a web platform for small-business teams to turn internal SOPs and other documents into structured onboarding. An owner creates a workspace, adds supported text files (TXT, Markdown, CSV or JSON), organizes material into training domains, and invites learners with a join code. Access requests remain pending until the owner approves them. Learners work through modules, quizzes and practice sandboxes, then can ask questions about workspace documents. The planned retrieval-augmented assistant is designed to show the passages used for each answer so learners can check the source. The current prototype includes Culinary SOPs, Tech Onboarding and Customer Support, with 30 clearly labeled sample SOPs. The web application uses React, Express and PostgreSQL and is hosted on Replit. A separate RunPod Serverless worker is intended to index documents and support retrieval, cited chat and optional LoRA fine-tuning. The team applied workshop skills in GenAI-assisted development, database-backed workspaces and cloud deployment.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

WEB APP (Replit): React 19 + Vite 8 + Tailwind CSS 4

Owner / Learner browser

| Clerk authentication

v

Express 5 API <----> PostgreSQL

|

+---- planned RunPod Serverless worker

text chunking -> all-MiniLM-L6-v2 embeddings -> HNSWlib retrieval

-> Qwen2.5-0.5B-Instruct cited responses; optional PEFT LoRA

The database stores workspaces, memberships, training content, progress, quiz results and messages. The worker path is prepared in code.

Models shown are configured defaults.

Replit for development / web hosting;

RunPod for the planned GPU-compute path.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We framed the problem around two roles: owners who maintain approved knowledge and new team members who need to learn it. The prototype turns that into an owner workflow (create a workspace, add materials, invite and approve learners) and a learner workflow (complete modules, quizzes and practice, then ask source-grounded questions). We iterated on role permissions, document revisions and clear unavailable states rather than presenting demo telemetry as real results. Thirty labeled sample SOPs support demonstration across three domains. Automated tests cover access isolation, approval, content revisions, progress, quizzes and worker failure states; they are engineering tests, not external user research. Phase 2 should validate usability and onboarding time with pilot learners. Replit hosts the web application; the RunPod worker must be built and GPU-tested before its AI path is treated as deployed.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4 - increase relevant skills for employment and decent jobs. SDG 8.2 - improve productivity through technology upgrading and innovation. SDG 9.b - support domestic technology development and innovation. Link's contribution is indirect: structured, searchable workplace learning and a locally built digital training prototype. These are intended pathways, not measured SDG outcomes.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The OECD 2025 D4SME Survey reports that 13% of surveyed SMEs used formal digital-skills training; 35% used internet research and 23% colleague knowledge-sharing.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary: median minutes from invitation to completion of required onboarding modules. Secondary: module completion rate (%) and pre/post quiz score. Track pilot sample size and results by team.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 pilot target : test with at least 10 learners; reduce median onboarding time by 20% versus each participating team's pre-Link baseline and achieve 80% completion of required modules. Record the baseline and results. Monitor compute use; do not claim carbon or waste savings without measurement.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [X] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [X] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [X] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [X] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19; Vite 8; Tailwind CSS 4	Web UI and build tools	MIT; ISC
Python worker libraries	FastAPI, PyTorch, Transformers, Sentence Transformers, HNSWlib, PEFT	MIT / BSD / Apache-2.0; per-package
Express 5; pg; Clerk SDKs	API, PostgreSQL access and auth SDKs	MIT; Clerk service terms
Qwen2.5-0.5B-Instruct	Configured pre-trained language model	Apache-2.0
all-MiniLM-L6-v2	Configured pre-trained embedding model	Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24th October 2026